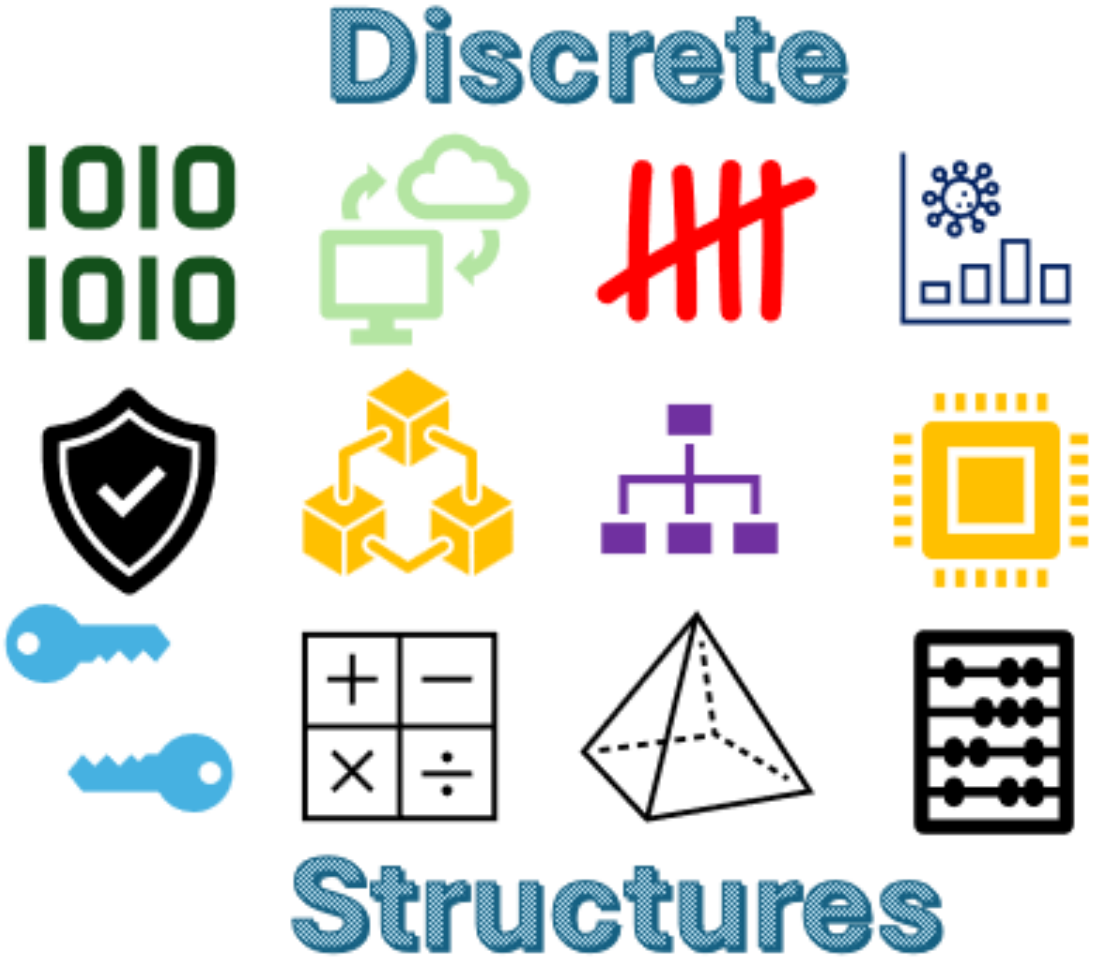
COMP 2310: Discrete Structures
Fall 2026

Chapter 4: Relation and digraph



Operations on relations: Complementary relation

- Let A = {1, 2, 3, 4} and B = {a, b, c}. Let

  R = {(1,a), (1, b), (2, b), (2, c), (3, b), (4,a)}

  S = {(1, b), (2, c), (3, b), (4, b)}.

- **Compute**
  - $\bar{R}$ (Complementary relation)
    - The complement of the relation R (denoted as $\bar{R}$) is defined as the set of all pairs in A x B that are not in R
      - $\bar{R}$ **= A x B – R**

        = {(1, a), (1, b), (1, c), (2, a), (2, b), (2, c), (3, a), (3, b), (3, c), (4, a), (4, b), (4, c)} - ***{(1,a), (1, b), (2, b), (2, c), (3, b), (4,a)}***

        = {(1, c), (2, a), (3, a), (3, c), (4, b), (4, c)}

Operations on relations: Inverse relation

- Let A = {1, 2, 3, 4} and B = {a, b, c}. Let

  $R = \{(1,a), (1, b), (2, b), (2, c), (3, b), (4,a)\}$

  $S = \{(1, b), (2, c), (3, b), (4, b)\}$.

- **Compute**
  - R^{-1} (Inverse relation)
  - $R^{-1} = \{(a, 1), (b, 1), (b, 2), (c, 2), (b, 3), (a, 4)\}$

Operations on relations: matrices

Let $A = \{1, 2, 3\}$ and let R and S be relations on A. Suppose that the matrices of R and S are

$$\mathbf{M}_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{M}_S = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

$$\mathbf{M}_{\overline{R}} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Flip the element to indicate the value is not present

$$\mathbf{M}_{R^{-1}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

Swap the rows and columns

$$\mathbf{M}_{R^{-1}} = (\mathbf{M}_R)^T$$

$$\mathbf{M}_{R\cap S} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{M}_{R\cap S} = \mathbf{M}_R \wedge \mathbf{M}_S$$

$$\mathbf{M}_{R\cup S} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\mathbf{M}_{R\cup S} = \mathbf{M}_R \vee \mathbf{M}_S$$

Operations on relations and resulting properties

Operations on relation exhibits several properties including:

- Reflexive
- Symmetric
- Transitive

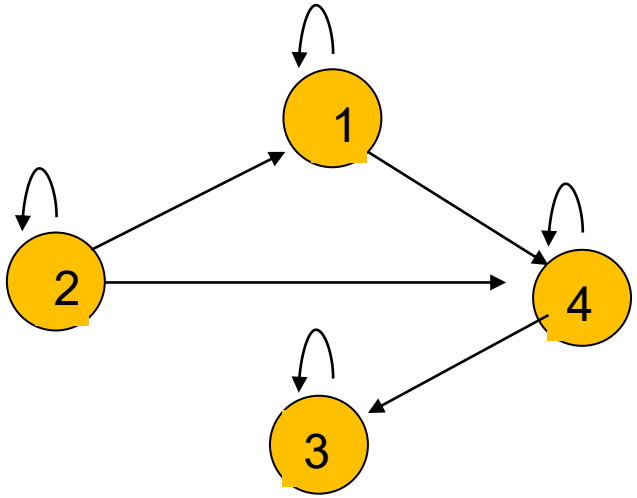
Properties of relations: looking back

Operations on relation exhibits several properties including:

- Reflexive
- Symmetric
- Transitive

Looking back: Properties of relations: reflexive

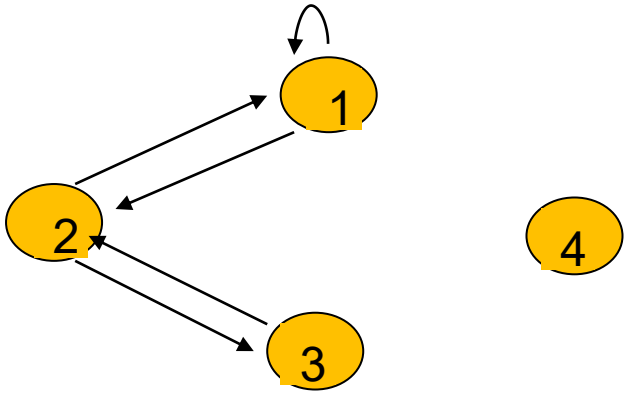
- A relation Q on set A, is reflexive if for all $a \in A$, then $(a, a) \in Q$
- Example: Let $A=\{1, 2, 3, 4\}$
  - $Q=\{ (1,1), (1,4), (2,1), (2,2), (2,4), (3,3), (4,3), (4,4) \}$
- ***Must have self pairs for all elements***



- A relation Q on set A, is **irreflexive** if no element relates to itself

Looking back: Properties of relations: symmetric

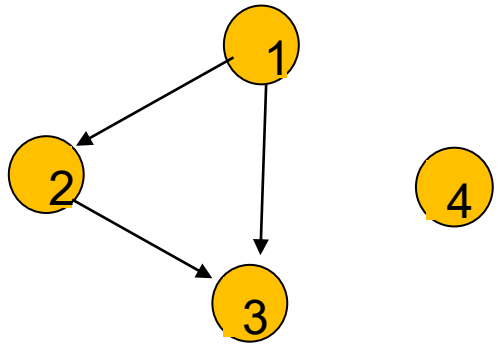
- A relation Q on set *A*, is symmetric
  - If $(a, b) \in Q$ then (b, a) is also $\in Q$
- $A = \{1, 2, 3\}$
  - $Q = \{ (1, 1), (1, 2), (2, 1), (2, 3), (3, 2) \}$



- A relation Q on set *A*, is **asymmetric** if whenever $(a, b) \in Q$, $(b, a) \notin Q$
  - If the reverse pair is not available

Looking back: Properties of relations: transitive

- A relation Q on set A, is transitive if $(a, b) \in Q$ and $(b, c) \in Q$ then $(a, c) \in Q$
- Example: Let A={ 1, 2, 3 } R={ (1, 2), (2, 3), (1, 3) }
- If an element a relates to b (written as (a,b)∈Q) and that element b relates to c (written as (b,c)∈Q, then the **first element** a should also **directly** relate to c (i.e., (a,c)∈Q).
- If your friend A is related to B and you as C are also related to B



Operations on relations and resulting properties: **reflexivity**

- Let A = {1, 2, 3} and consider two reflexive relationships
  - R = {(1, 1), (1, 2), (1, 3), (2, 2), (3, 3)}
  - S = {(1, 1), (1, 2), (2, 2), (3, 2), (3, 3)}
    - $\boldsymbol{R^{-1}}$ = {(1, 1), (2, 1), (3, 1), (2, 2), (3, 3)}; R and R^{-1} are both reflexive.
    - $\overline{\boldsymbol{R}}$ = {(2, 1), (2, 3), (3, 1), (3, 2)} ← (A x A – R) is irreflexive while R is reflexive
    - $\boldsymbol{R \cap S}$ = {(1, 1), (1, 2), (2, 2), (3, 3)} is reflexive
    - $\boldsymbol{R \cup S}$ = {(1, 1), (1, 2), (1, 3), (2, 2), (3, 2), (3, 3)} is reflexive.

Operations on relations and resulting properties: **Symmetric**

- Let A = {1, 2, 3} and consider two symmetric relationships
  - R = {(1, 1), (1, 2), (2, 1), (1, 3), (3, 1)}
  - S = {(1, 1), (1, 2), (2, 1), (2, 2), (3, 3)}.

    - $\boldsymbol{R^{-1}}$ = {(1, 1), (2, 1), (1, 2), (3, 1), (1, 3)} ; R and R^{-1} are both symmetric.
    - $\boldsymbol{\bar{R}}$ = {(2, 2), (2, 3), (3, 2), (3, 3)} R and $\bar{R}$ are both symmetric
    - $\boldsymbol{R \cap S}$ = {(1, 1), (1, 2), (2, 1)} is symmetric
    - $\boldsymbol{R \cup S}$ = {(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (3, 1), (3, 3)} is symmetric

Operations on relations and resulting properties: **Transitivity**

- **Intersection:** If R and S are transitive, R ∩ S is transitive
- **Inverse**: If R and S are transitive, R^{-1} U S^{-1} are not transitive
- **Union:** If R and S are transitive, R U S is not transitive
- **Complement:** If R and S are transitive, $\bar{R}$ **and** $\bar{S}$ **are not transitive**

Operations on relation: **Composition**

- (S ◦ R)(A1) = S(R(A1))
- **In other words:** S∘R={(a,c)| there exists b∈A such that (a,b)∈R and (b,c)∈S}
- Example:
- A = {1, 2, 3, 4}
- R = {(1, 2), (1, 1), (1, 3), (2, 4), (3, 2)}
- S = {(1, 4), (1, 3), (2, 3), (3, 1), (4, 1)}
  - S ◦ R = {(1, 4), (1, 3), (1, 1), (2, 1), (3, 3)}
  - How?
    - Check each element of in R. For instance, (1, 2) where a = 1, b = 2.
    - Find a pair in S where, a in S = b in R. For instance, (2, 3) in S
    - Therefore, (a, c) = (1, 3) ∈ S∘R
    - Repeat for other pairs in R

The rule: Take each pair in R, look at its second number, and find a pair in S whose first number matches it.

Examples

- A = {a, b, c}; B = {1, 2, 3}
  R = {(a, 1), (b, 1), (c, 2), (c, 3)}
  S = {(a, 1), (a, 2), (b, 1), (b, 2)}

  Compute **(a)** $\overline{\mathbf{R}}$; **(b)** R ∩ S; **(c)** R ∪ S; **(d)** S^{-1}
  Answers: Discussed in the class. Also see previous examples

- Let A = {1, 2, 3, 4}. Let
  R = {(1, 1), (1, 2), (2, 3), (2, 4), (3, 4), (4, 1), (4, 2)}
  S = {(3, 1), (4, 4), (2, 3), (2, 4), (1, 1), (1, 4)}

  1. Compute S ◦ R (S composed with R) start checking with R set
     {(1, 1), (1, 4), (1, 3), (2, 1), (2, 4), (3, 4), (4, 1), (4, 4), (4, 3)}
  2. Compute R ◦ S (R composed with S) start checking with S set
     {(3, 1), (3, 2), (4, 1), (4, 2), (2, 4), (2, 1), (2, 2), (1, 1), (1, 2)}